

## OS & SP

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### SKILL – 03

1.



```
GNU nano 8.7.1 skill3.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_LENGTH 100

int main() {
    char history[MAX_HISTORY][MAX_LENGTH];
    char input[MAX_LENGTH];
    int count = 0;
    int current = -1;

    printf("=== Command History Program ===\n");
    printf("Enter commands (type 'exit' to stop)\n\n");

    while (1) {
        printf("Command: ");
        fgets(input, MAX_LENGTH, stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (count < MAX_HISTORY) {
            strcpy(history[count], input);
            count++;
        } else {
            for (int i = 1; i < MAX_HISTORY; i++)
                strcpy(history[i - 1], history[i]);

            strcpy(history[MAX_HISTORY - 1], input);
        }

        current = count;

        printf("Command stored successfully.\n");
    }

    printf("\n=== Command History ===\n");

    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
    }

    printf("\nEscape Sequence Demonstration:\n");
    printf("New Line -> Hello\nWorld\n");
    printf("Tab -> Hello\tWorld\n");
    printf("Backslash -> C:\\Program Files\n");

    return 0;
}
```

## Output :

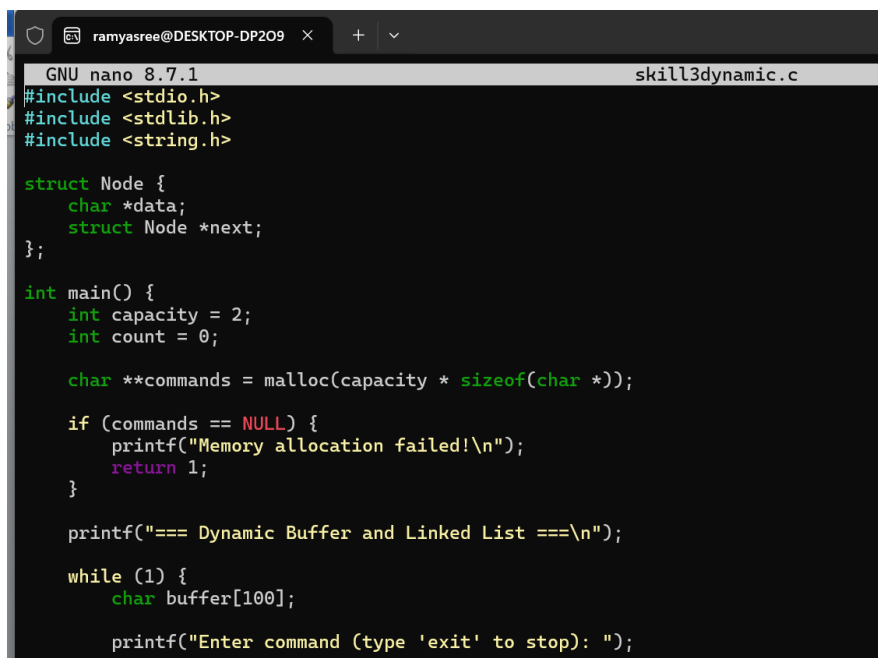
```
ramyasree@DESKTOP-DP209U6:~$ mkdir Skill
mkdir: Skill: File exists
ramyasree@DESKTOP-DP209U6:~$ cd Skill
ramyasree@DESKTOP-DP209U6:~/Skill$ nano skill3.c
ramyasree@DESKTOP-DP209U6:~/Skill$ gcc history.c -o history
cc1: fatal error: history.c: No such file or directory
compilation terminated.
ramyasree@DESKTOP-DP209U6:~/Skill$ gcc skill3.c -o skill3
ramyasree@DESKTOP-DP209U6:~/Skill$ ./skill3
=== Command History Program ===
Enter commands (type 'exit' to stop)

Command: ls
Command stored successfully.
Command: pwd
Command stored successfully.
Command: history
Command stored successfully.
Command: exit

=== Command History ===
1: ls
2: pwd
3: history

Escape Sequence Demonstration:
New Line -> Hello
World
Tab -> Hello    World
Backslash -> C:\Program Files
ramyasree@DESKTOP-DP209U6:~/Skill$ nano skill3.c
ramyasree@DESKTOP-DP209U6:~/Skill$ |
```

## 2.



```
GNU nano 8.7.1 skill3dynamic.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Node {
    char *data;
    struct Node *next;
};

int main() {
    int capacity = 2;
    int count = 0;

    char **commands = malloc(capacity * sizeof(char *));

    if (commands == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }

    printf("=== Dynamic Buffer and Linked List ===\n");

    while (1) {
        char buffer[100];

        printf("Enter command (type 'exit' to stop): ");
```

```
ramyasree@DESKTOP-DP209  ×  +  ∨
GNU nano 8.7.1 skill3dynamic.c
char buffer[100];

printf("Enter command (type 'exit' to stop): ");
fgets(buffer, sizeof(buffer), stdin);

buffer[strcspn(buffer, "\n")] = '\0';

if (strcmp(buffer, "exit") == 0)
    break;

if (count == capacity) {
    capacity *= 2;

    char **temp = realloc(commands,
                           capacity * sizeof(char *));

    if (temp == NULL) {
        printf("Memory reallocation failed!\n");

        for (int i = 0; i < count; i++)
            free(commands[i]);

        free(commands);
        return 1;
    }
}
```

```
ramyasree@DESKTOP-DP209U6:~/Skill$ nano skill3dynamic.c
ramyasree@DESKTOP-DP209U6:~/Skill$ gcc skill3dynamic.c -o skill3dynamic
ramyasree@DESKTOP-DP209U6:~/Skill$ ./skill3dynamic
=== Dynamic Buffer and Linked List ===
Enter command (type 'exit' to stop): ls
Enter command (type 'exit' to stop): pwd
Enter command (type 'exit' to stop): gcc test.c
Buffer resized to 4 elements.
Enter command (type 'exit' to stop): exit

=== Stored Commands ===
1. ls
2. pwd
3. gcc test.c

=== Linked List ===
ls -> pwd -> gcc test.c -> NULL

All dynamically allocated memory released.
ramyasree@DESKTOP-DP209U6:~/Skill$ |
```